

Results

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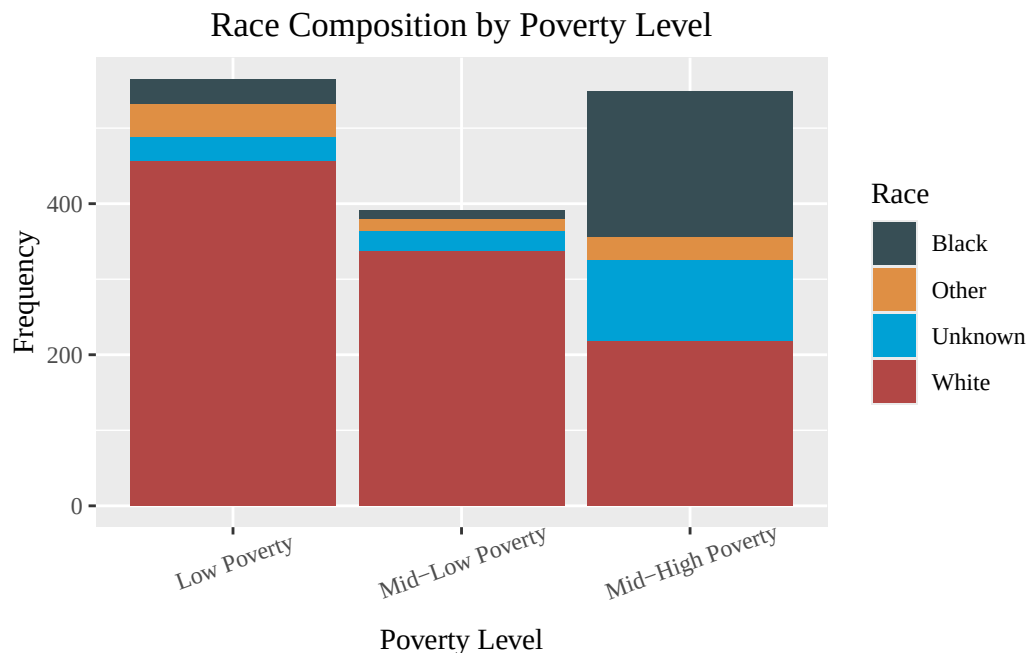
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1 Sample Characteristics and Demographics

The sample consisted of 1504 students across 19 classrooms in 9 Indiana secondary schools. Respondents were primarily students early in their secondary education, with the vast majority of respondents being ninth-grade students (84%; $n = 1265$), with smaller cohorts of grades 10 ($n = 82$), 11 ($n = 16$), and 12 ($n = 21$) also included. Grade level was not reported by 120 students. Correspondingly, the age distribution was centered at age 14 ($n = 976$), followed by 15 ($n = 299$), 13 ($n = 157$), 16 ($n = 43$), 17 ($n = 22$) and 18 ($n = 7$).

Respondents were distributed evenly between sexes, with 765 reporting female and 747 reporting male. The majority of respondents were White ($n = 1011$), followed by Black ($n = 237$), Asian ($n = 49$), Native American ($n = 33$), and Native Hawaiian ($n = 8$). Due to small sample sizes, Asian, Native American, and

Native Hawaiian respondents were combined into the “Other Race” category. 166 respondents did not report race. Notably, 64% of respondents come from schools associated with higher socioeconomic status, either belonging to the “low poverty” or “mid-low” poverty group, which may limit the generalizability of any findings. The “mid-high” poverty group contained significantly higher proportions of students who reported their race as black or unknown.



2 Descriptive Statistics

2.1 Distributions of Test Scores

Assessment scores varied across standards. Standards 2 and 3 were approximately normally distributed, while Standard 4 was significantly skewed left, with most students scoring in the upper range. Standard 4 had the highest median score (71.4%), followed by the overall score (61.5%), Standard 2 (57.1%), and Standard 3 (50%). Visual inspection of distribution highlights the disparity in number of assessment items across standards (21 items for Standard 2, 4 for Standard 3, 14 for Standard 4), which affects the resolution and potentially the reliability of each distribution.

Distribution of Test Scores by Standard

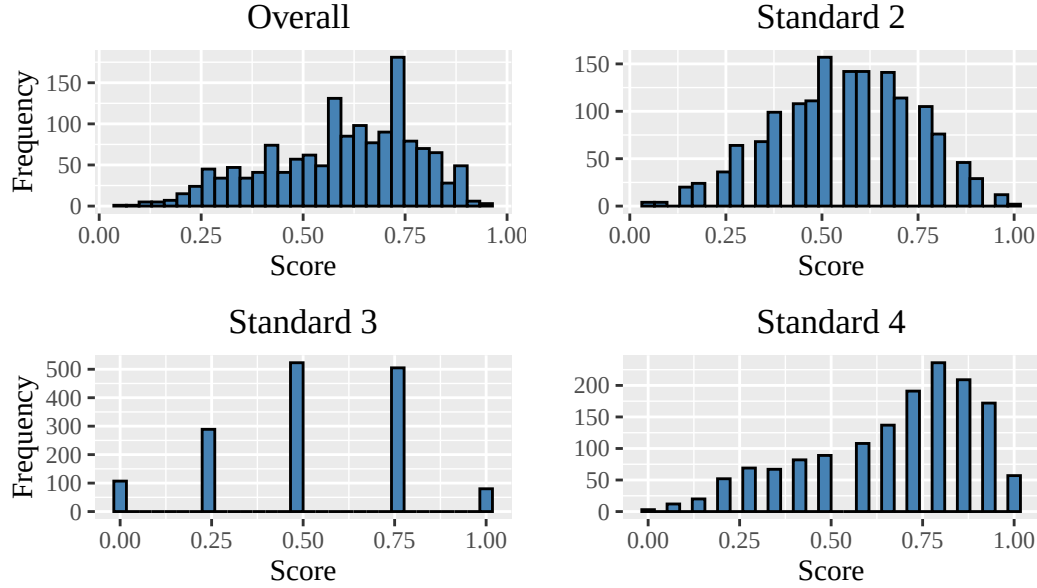


Table 1: Summary Statistics for Test Scores

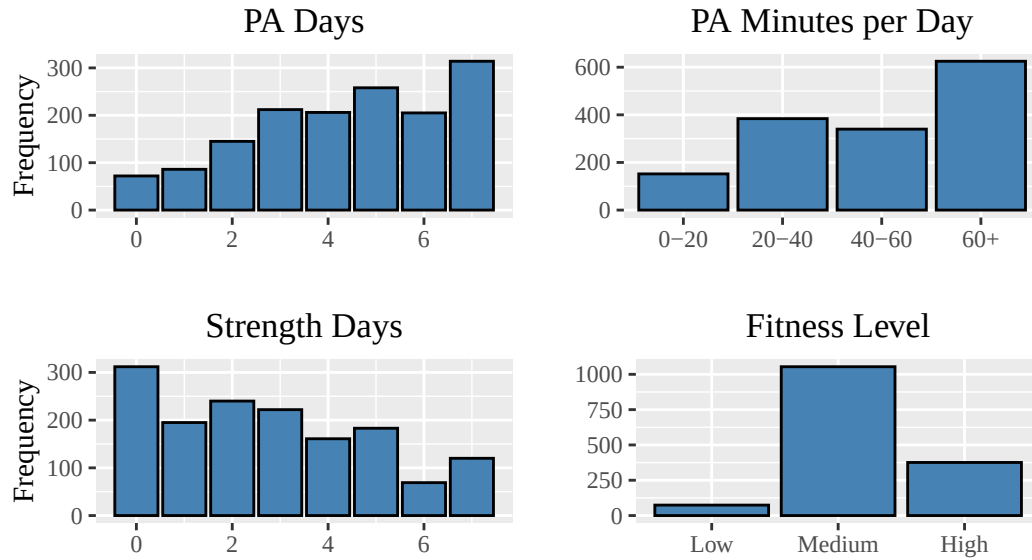
Standard	Mean	Q1	Median	Q3
Overall	0.595	0.487	0.615	0.744
Standard 2	0.560	0.429	0.571	0.714
Standard 3	0.527	0.250	0.500	0.750
Standard 4	0.669	0.500	0.714	0.857

2.2 Distributions of PA Variables

Students generally reported high levels of physical activity. The distribution of PA Days per Week was left-skewed, with a median of 5 days and a most common response of 7 days per week. Similarly, PA Minutes per Day was left skewed, with the most common response being 60+ minutes per day. Strength Days showed a different pattern, with a right-skewed distribution, a median of 3 days, and a most common response of 0 days.

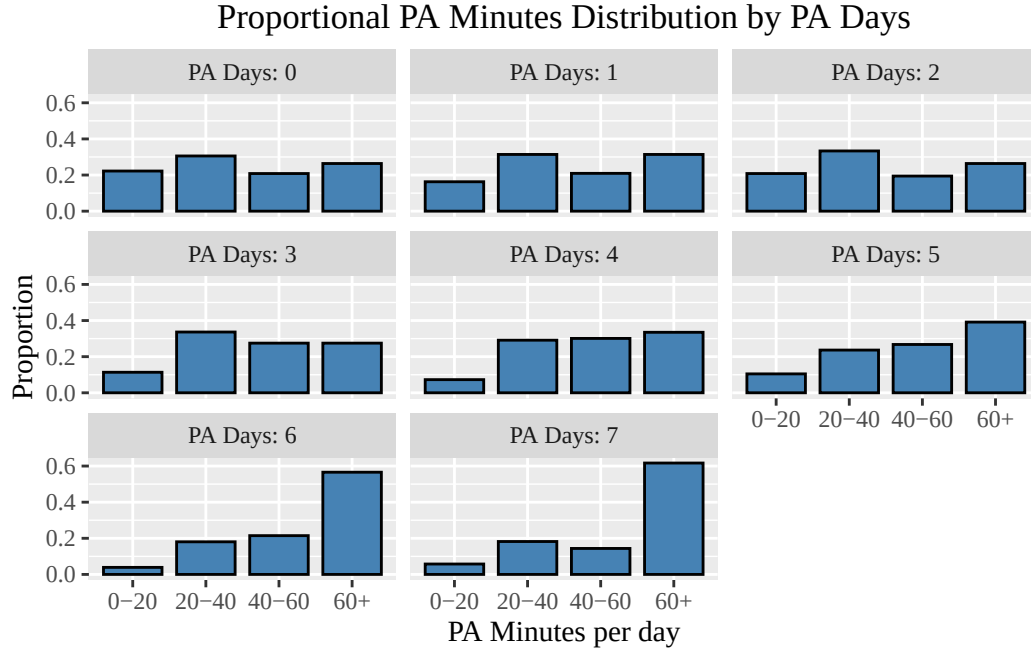
The majority of students fell into the “Medium Fitness” cohort ($n = 1054$). A substantial number of students belonged to the “High Fitness” cohort ($n = 376$), though only a small number of students belonged to the “Low Fitness” cohort ($n = 74$). The low sample size of the “Low Fitness” category limits the statistical power of analyses involving this group.

Physical Activity Metric Distributions



2.3 PA Days and PA Minutes per Day Inconsistency

Students generally reported high levels of physical activity. The median reported days of physical activity was 5, and most students reported 40+ minutes per day of physical activity. There was, however, a notable discrepancy between these statistics. A student who reports an average of 60+ minutes of physical activity per day should also report 7 days of physical activity per week, by definition. While students did report higher average minutes of daily physical activity as they reported higher days of physical activity per week, there was inconsistency in how these metrics were reported. For example, 19 students reported both 0 days of physical activity per week, and an average of 60+ minutes of physical activity per day. The plot of the distribution of PA Minutes per Day by reported PA Days per Week makes this inconsistency clear.



3 Exploratory Findings

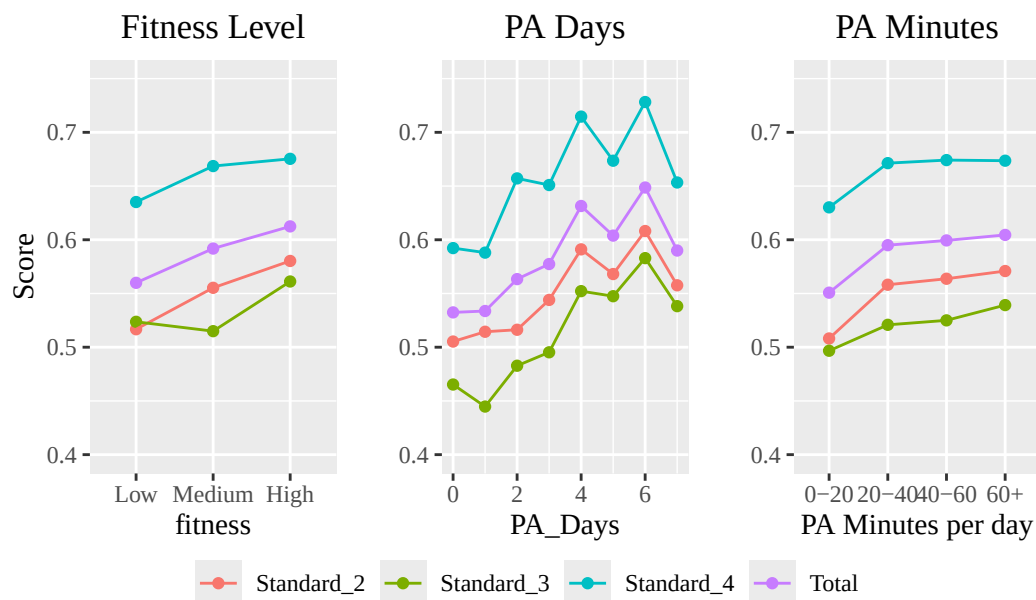
3.1 Correlation Between Scores and Outcome Variables

Assessment scores generally correlated positively with physical activity across all three outcome metrics. Students in the High Fitness cohort scored higher than the Medium Fitness cohort, which scored higher than the Low Fitness cohort across most standards. The one exception being Standard 3, where the Low Fitness cohort outperformed Medium Fitness. The small sample size of the Low Fitness cohort ($n = 74$) should be considered when interpreting this result.

A similar positive relationship was observed between test scores and PA Days per Week. Students who reported higher numbers of physically active days per week also tended to score higher across each standard. Unexpectedly, there is a marked decline in scores between students who report 6 and 7 PA Days per Week.

The relationship between PA Minutes per Day and test scores was also positive across each standard. There is a notable jump in scores between students who report less than 20 minutes per day and students who report 20-40 minutes, with scores remaining stable across the higher activity categories. This pattern motivated the dichotomization of this variable, as mentioned in the Methods section.

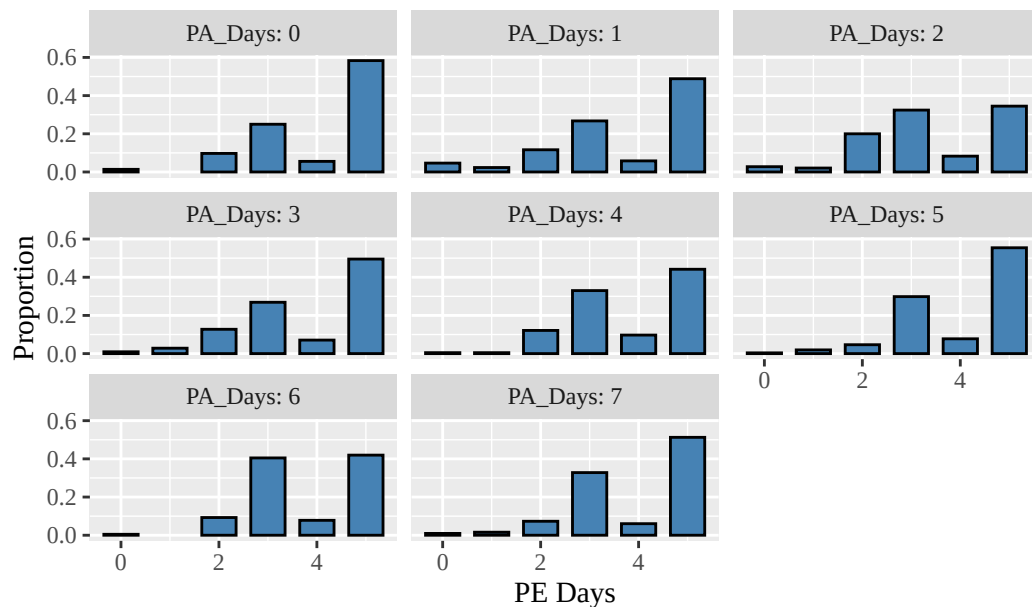
Correlation Between Scores and PA Metrics



3.2 PE days and PA Days

To examine whether the number of PE Days per Week was associated with activity levels, we compared the distributions of PE Days across each level of PA Days per Week. Students tend to report either 3 or 5 days per week of PE classes, with no discernible pattern across reported days of physical activity. While a chi-square test indicated a statistically significant association ($\chi^2 = 76.57, p < 0.001$), the effect size was negligible (Cramér's $V = 0.101$). This suggests that the raw number of days of PE per week has little impact on the number of days that a student reports being physically active.

Distribution of PE Days by PA Days



4 Model Results

4.1 PA Minutes Threshold

A logistic regression was fit to examine the association between the overall standardized assessment scores and the probability of a student exceeding the sedentary threshold, as measured by PA Minutes per Day. The standardized, overall assessment score was statistically significant ($p = 0.001$), which indicates that higher scores are associated with larger odds of a student exceeding the 20-minute activity threshold. However, this model offers little explanatory power, with an AUC of 0.586, and McFadden's R^2 of 0.01. No overdispersion was detected in this model.

When standardized assessment scores separated by standard were modeled against the same sedentary threshold variable, only Standard 2 had a significant predictor ($p = 0.004$). Standards 3 and 4 were not significant predictors ($p = 0.893, p = 0.634$, respectively). Model fit remained poor with McFadden's $R^2 = 0.014$ and an AUC of 0.590, consistent with the overall score model.

When demographic covariates were added, the overall score was only marginally significant ($p < 0.1$) and decreased in magnitude. Standard 2 assessment scores remained significant at the $p < .05$ threshold. Among the demographic variables, Sex was marginally significant ($p < 0.1$) and Poverty Level was highly significant ($p < 0.01$). Males were associated with higher odds of surpassing the sedentary threshold, and belonging to the "Mid-High Poverty" class was negatively associated with active behavior, compared to the baseline "Low Poverty" class. Model fit remained modest across all four models, with McFadden's R^2 increasing to 0.034 and AUC to 0.643 with the addition of demographic variables.

Table 2: Logistic Regression: PA Minutes Threshold

	<i>Dependent variable:</i>			
	Active Daily (>20 min)			
	(1)	(2)	(3)	(4)
Overall Score	0.267*** (0.083)		0.171* (0.093)	
Standard 2		0.346*** (0.120)		0.299** (0.123)
Standard 3		0.013 (0.098)		−0.082 (0.103)
Standard 4		−0.058 (0.121)		−0.061 (0.124)
Sex			0.342* (0.177)	0.339* (0.177)
Mid-Low Poverty			−0.039 (0.255)	−0.023 (0.256)
Mid-High Poverty			−0.732*** (0.237)	−0.749*** (0.237)
Other Race			0.016 (0.420)	0.046 (0.422)
Unknown Race			0.078 (0.317)	0.086 (0.318)
White			0.084 (0.250)	0.107 (0.251)
Latino			0.138 (0.272)	0.127 (0.273)
Constant	2.212*** (0.088)	2.223*** (0.089)	2.288*** (0.285)	2.288*** (0.286)
McFadden's R ²	0.010	0.014	0.031	0.035
AUC	0.586	0.590	0.639	0.643
Observations	1,501	1,501	1,495	1,495
Log Likelihood	−486.983	−485.356	−476.129	−474.148
Akaike Inf. Crit.	977.966	978.711	970.258	970.297

Note:

*p<0.1; **p<0.05; ***p<0.01

4.2 PA Days per Week

As overdispersion was detected in binomial models, a quasi-binomial regression model was fit to examine the association between the same assessment scores and physical activity, this time measured by self-reported physically active days per week. When fit as the sole predictor, overall score was highly significant ($p < 0.01$). Scoring higher overall on the knowledge-based PE exam is associated with an increased probability that a student will be active on a given day.

When evaluating the association between individual standards and PA Days, both Standard 2 and Standard 3 were significant ($p < 0.05$, $p < 0.01$, respectively). Standard 4 was not significant, and its coefficient was even slightly negative. The magnitude of the association of Standard 3 was slightly larger than that of Standard 2.

When demographic variables were added, overall scores remained a significant predictor, as well as Standard 2 and Standard 3 assessment scores. Sex, poverty level, and whether or not a student was latino were all significant predictors. Males were associated with higher probabilities of physical activity, and Latino students were negatively associated with probability of physical activity. Notably, belonging to the “Mid-Low Poverty” class was associated with higher probabilities of physical activity, higher than both the “Mid-High Poverty” and “Low Poverty” classes.

Table 3: Binomial Regression: PA Days per Week

	<i>Dependent variable:</i>			
	PA Days per Week (0-7)			
	(1)	(2)	(3)	(4)
Overall Score	0.150*** (0.032)		0.152*** (0.036)	
Standard 2		0.105** (0.045)		0.113** (0.045)
Standard 3		0.125*** (0.038)		0.090** (0.039)
Standard 4		−0.024 (0.047)		−0.003 (0.047)
Sex			0.510*** (0.066)	0.502*** (0.066)
Mid-Low Poverty			0.338*** (0.085)	0.329*** (0.086)
Mid-High Poverty			0.036 (0.088)	0.050 (0.088)
Other Race			−0.158 (0.160)	−0.169 (0.161)
Unknown Race			−0.111 (0.129)	−0.126 (0.129)
White			0.101 (0.103)	0.091 (0.103)
Latino			−0.308*** (0.104)	−0.311*** (0.104)
Constant	0.513*** (0.032)	0.514*** (0.033)	0.162 (0.110)	0.173 (0.111)
Observations	1,498	1,498	1,495	1,495
<i>Note:</i>		*p<0.1; **p<0.05; ***p<0.01		

4.3 Fitness Cohorts

A logistic regression was fit to model the relationship between assessment scores and the probability of students meeting physical activity recommendations, and thus belonging to the “High Fitness” cohort. When fit as the only predictor, overall score had a significant ($p < 0.05$) positive relationship with the probability that a student belongs to the “High Fitness” cohort. The model offers poor explanatory power (McFadden’s

$R^2 = 0.003$, $AUC = 0.542$).

With each standard included separately, the model is similarly poor-fitting (McFadden's $R^2 = 0.009$, $AUC = 0.570$). Standard 2 and Standard 3 have significant ($p < 0.05$, $p < 0.01$, respectively) positive associations with the probability of belonging to the “High Fitness” cohort. Standard 4 is also significant ($p < 0.05$), but the relationship is negative. This result is surprising, but should be viewed with the context of poor overall model fit.

When demographic variables are added, the resulting models have significantly improved fits (McFadden's $R^2 = 0.039$, $AUC = 0.635$), as one would expect. Overall score had its significance level attenuated ($p < 0.1$). Standard 2 maintained the same level of significance ($p < 0.05$), Standard 3 saw its significance attenuated ($p < 0.1$), and while Standard 4 maintains a negative coefficient, the relationship is not significant.

Table 4: Logistic Regression: Fitness Cohort

	<i>Dependent variable:</i>			
	Meets Fitness Recommendations			
	(1)	(2)	(3)	(4)
Overall Score	0.128** (0.061)		0.132* (0.068)	
Standard 2		0.205** (0.083)		0.210** (0.086)
Standard 3		0.198*** (0.071)		0.122* (0.073)
Standard 4		−0.198** (0.087)		−0.146 (0.090)
Sex			0.647*** (0.126)	0.628*** (0.126)
Mid-Low Poverty			0.551*** (0.149)	0.534*** (0.150)
Mid-High Poverty			−0.397** (0.175)	−0.375** (0.176)
Other Race			−0.426 (0.330)	−0.449 (0.331)
Unknown Race			−0.067 (0.259)	−0.089 (0.259)
White			−0.195 (0.206)	−0.207 (0.207)
Latino			0.167 (0.203)	0.162 (0.204)
Constant	−1.098*** (0.060)	−1.109*** (0.060)	−1.327*** (0.221)	−1.313*** (0.222)
McFadden's R ²	0.003	0.009	0.036	0.039
AUC	0.542	0.570	0.624	0.635
Observations	1,499	1,499	1,494	1,494
Log Likelihood	−842.062	−836.134	−811.535	−808.435
Akaike Inf. Crit.	1,688.123	1,680.269	1,641.070	1,638.869

Note:

*p<0.1; **p<0.05; ***p<0.01

5 Supplementary Findings

5.1 Sports Participation by SES

5.2 Fitness Cohort Comparisons

5.3 Other Variable Distributions (weight status, weight action, computer/tv hours)